# Christos Giannakopoulos

Data Scientist
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Personal Webpage

## PROFESSIONAL SUMMARY

Data Scientist with a Physics Ph.D. and **5+ years** of hands-on experience building and deploying predictive models, designing **end-to-end analysis pipelines**, and mining **unstructured data** using advanced statistical, and machine learning techniques including **natural language processing (NLP)** for topic detection, and keyword extraction. Proven ability to independently drive analytics projects from definition to solution and implementation on **distributed computing** environments while translating complex findings into clear, actionable recommendations for non-technical stakeholders.

## TECHNICAL SKILLS

**Programming:** Python (5+ yrs), SQL (2+ yrs), MATLAB (6+ years), Bash
**ML & Data Science:** scikit-learn, TensorFlow, Keras, NumPy, Pandas, SciPy, Matplotlib; supervised & unsupervised ML, feature engineering
**NLP:** spaCy, KeyBERT/BERTopic, LLM fine-tuning, Agentic RAG
**Distributed / Big Data:** HPC/SLURM clusters, multi-terabyte data pipelines; PySpark (familiar)
**MLOps & Dev Tools:** Git/GitHub, Docker, Linux, end-to-end pipeline orchestration

## EXPERIENCE

**Research Data Scientist** — *University of Cincinnati* 2021 – 2026

Ph.D. student researcher embedded in a multi-institution, federally funded physics program. Owned analytics and codebase infrastructure for a 50+ person collaboration with deliverables reviewed by program leadership.

- Built an end-to-end **NLP pipeline** using spaCy to mine **large-scale, multi-source unstructured data**, enabling **named entity recognition (NER)** and **topic detection**, reducing manual review time by **35%**.
- Conducted extensive **exploratory data analysis** on high-dimensional, noisy sensor datasets; created new input **features** through domain-informed **feature engineering** and dimensionality reduction (PCA/SVD), extracting insights that improve model confidence by **95%**.
- Built and optimized **predictive models** on complex, multi-modal datasets (time series, point clouds, spatial maps) using advanced mathematical and statistical techniques (Gaussian processes, Bayesian inference), improving detection sensitivity by a factor of **2×**.
- Maintained large-scale data pipelines processing **multi-terabyte datasets** on **distributed computing clusters**, delivering production-quality analytical outputs under strict deadlines.
- Designed and executed novel data acquisition strategies; built statistical validation tools that cross-validated model outputs against independent measurements, meeting analysis deadlines **50% sooner** and enabling **data-driven decisions** for program stakeholders.
- Developed model validation frameworks to quantify previously unmeasured model uncertainties, benchmarking **end-to-end system performance** and informing hardware design trade-offs, resulting in a **78% reduction in compute costs**.
- Led a **cross-functional team** on a six-month project resulting in published deliverables that became the **program-wide analysis standard**.
- Designed scalable algorithms improving pipeline runtime by **8×**, enabling near real-time data quality monitoring across distributed infrastructure (high performance compute clusters).
- Architected a **self-correcting RAG** custom agent leveraging the 8B Llama 3.1 LLM to provide domain-specific Q&A.
- Served as **subject matter expert** for data system performance and model validation and mentored junior researchers in statistical methods, analysis best practices, and ML workflows.

**Teaching Assistant** — *University of Cincinnati* 2019 – 2021

- Instructed 200+ students while implemented active-learning techniques that improved engagement and outcomes by **78%**. Recognized for translating complex technical material into clear, accessible explanations.

## EDUCATION

**Ph.D. in Physics — University of Cincinnati** 2026

**B.S. in Physics — Hillsdale College** 2019

## CERTIFICATIONS & TRAINING

**Machine Learning Specialization — DeepLearning.AI**
3-course specialization covering Neural Networks, CNNs, Random Forests, Reinforcement Learning, Anomaly Detection, Clustering, and Recommender Systems.

**Experimental Systems Engineering — Harvard University & Caltech**
Designed and characterized precision measurement systems; performed end-to-end hardware validation and statistical error analysis for a next-generation sensor array.

## AWARDS & PUBLICATIONS

**"Calibration Measurements of the BICEP3 and BICEP Array CMB Polarimeters from 2017 to 2024"** Christos Giannakopoulos — SPIE — 2024 *(predictive modeling, data analysis, statistical analysis, data mining)*

**Best Recitation Instructor — Exceptional Teaching Award** University of Cincinnati *2019, 2020*